

Measuring perturbation–reconvergence cost in MATSim for charging-infrastructure placement

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1 Motivation

Charging-infrastructure planning can be formulated as a bi-level leader–follower problem, where an infrastructure planner determines charger deployment decisions at the upper level, while transport users respond through routing and charging decisions at the lower level. In our setting, the upper-level solves the infrastructure configurations, including charger locations, capacities, and pricing, while MATSim evaluates the resulting transport response at the lower level. Each candidate configuration requires extensive iterative simulation and replanning until the system response is sufficiently converged. With on the order of 100 or more candidate evaluations, this makes the bi-level optimization computationally expensive, as each lower-level evaluation can itself require numerous MATSim iterations: reported costs for a single converged run range from 35–42 minutes on an HPC cluster (Natterer et al. 2026) to 4.2 days on a 28-core server (Patwary et al. 2021).

Fadranski et al. (2026) size charging infrastructure for electrified long-haul freight on German motorways with MATSim inside an NSGA-II search over charger locations, the same domain, simulator and decision variable as this work. Converged MATSim runs being expensive, they drove the search from a queueing surrogate built on a single-iteration run, then validated that shortcut against ten full 100-iteration replanning runs. Mean waiting times differed by up to 92% (59 to 67% for the balanced solutions), and failed trips fell from 5,511 (8%) to 393 (0.6%). The size of that difference is known because the authors ran the validation. Additionally, Patwary et al. (2025) sited public chargers on a 10% MATSim model of Montreal with a single-shot physical metamodel. It predicted a 21% reduction in peak-hour queueing against a random-assignment benchmark; run back through MATSim, the peak number of queued vehicles rose instead, which the authors attribute to demand elasticity the metamodel cannot represent. Both studies quantify the gap between surrogate prediction and simulator response by running the full simulation.

2 The gap

The classical bi-level literature largely either collapses an analytical equilibrium into a single reformulated program, leaving no equilibrium solves to count, or counts lower-level solves while treating each one as an exact oracle of fixed cost. In neither case does the cost of a solve depend on how far the scenario moved; that dependence only exists once the lower level is a black box that converges by iteration.

How many MATSim iterations a perturbed scenario needs to reach equilibrium again is understood qualitatively, but the computation costs of the simulation-based optimisation over MATSim are still based on assumptions. To the best knowledge of the authors, reconvergence cost has not been reported as a function of perturbation magnitude. We measure that function for charging-infrastructure placement on an electric road-freight scenario, report per KPI against a stated convergence criterion, and use a cold-run control to test whether warm-starting biases the answer.

3 How the field accelerates a simulation-based loop

Existing literature achieves the computational efficiency through an analytical equilibrium or, for derivative-free bi-level problems, by assuming a twice-differentiable, strongly convex lower level; MATSim offers neither. Table 1 groups the approaches by mechanism rather than by method name, so one method can appear in several rows.

The first mechanism is the most populated, and the methods in it generally presuppose that a single evaluation returns a trustworthy answer. The fourth mechanism substitutes the evaluation entirely; both studies above sit there, and the training cost is substantial before amortisation, roughly 13,800 MATSim runs for the cross-city surrogate of Natterer et al. (2026) and 600 for Roman et al. (2026).

The second mechanism keeps the simulator but makes each evaluation cheaper. Two methods formalise this: stochastic kriging and multi-fidelity methods. Stochastic kriging assumes the cheap evaluation’s error is zero-mean noise, which a truncated run violates: it is not noisy around equilibrium, it is systematically short of it. Multi-fidelity methods tolerate such bias, but only where it is stable across the design space. Whether truncation bias is stable under perturbation is, to the best knowledge of the authors, not reported. The fidelity dial used in MATSim work to date has been population sample size (Hörl 2022) rather than convergence depth.

The third mechanism carries state forward. Nicolai (2013) established warm and hot starting for MATSim, reporting 100, 60 and 30 iterations for cold, warm and hot starts of one Zurich scenario. Patwary et al. (2021) warm-start from a converged solution and then allow 50 internal iterations on their large network and 150 on their toy one; their objective is a fit to observed counts, so too short an inner budget shows up as a poor fit. Infrastructure design has no such external check, since there are no observed counts for a layout that does not yet exist. `opdyts` (Flötteröd 2017) is the most developed in this mechanism and the only method in the table that treats incomplete convergence as a first-class term, separating a Uniformity Gap that penalises stochastic noise from an Equilibrium Gap that penalises distance from equilibrium.

This paper’s experimental design follows its prescription for generating candidates: “*relatively small variations of the currently best decision variable*”, each started from “*the stationary state of the currently best decision variable*”. In the reported experiments where variation is instantiated at a single magnitude, one grid step of ± 25 cent and ± 30 minutes, with the seven experiments differing in candidate-set size M . Its dependence on variation magnitude is therefore left open, and that dependence is what this paper measures. The Equilibrium Gap weight in the accompanying bound is $v = \|c\|L/(1-L)$,

Table 1: Mechanisms for reducing the cost of a simulation-based optimisation over MATSim.

#	Mechanism	Exemplars
1	Search smarter: fewer evaluations	Bayesian optimisation (Koutounidis et al. 2025); trust-region metamodel (Patwary et al. 2021); NSGA-II (Fadranski et al. 2026); ranking and selection (Hong et al. 2021)
2	Run rougher: cheaper evaluations, same simulator	multi-fidelity methods (Peherstorfer et al. 2018); stochastic kriging (Ankenman et al. 2010); population down-sampling (Hörl 2022); fixed 50-iteration inner budget (Patwary et al. 2021)
3	Carry state forward: reused evaluations	warm and hot starts (Nicolai 2013); opdyts (Flötteröd 2017); warm-started calibration (Patwary et al. 2021)
4	Leave the simulator out: substituted evaluations	queueing surrogate (Fadranski et al. 2026); single-shot metamodel (Patwary et al. 2025); learned surrogates (Natterer et al. 2026; Roman et al. 2026)

with L the simulator’s contraction rate; the paper notes that the formulas rely on “*in general unknown problem properties* (simulator noise variances, Lipschitz constant)”.

4 Measuring it

A measured curve is directly consumable by three of those mechanisms: it is the cost-against-fidelity trade-off that truncation and multi-fidelity schemes presuppose; it says what warm-starting buys as a function of how far the scenario moved; and it supplies an empirical estimate of the Equilibrium Gap weight that opdyts currently self-tunes online.

Scenarios. Two, sharing the vehicle model, consumption map and run recipe. Brandenburg motorways from the MATSim electric-vehicle contrib, 39,275 links and 24,432 nodes, carries 250 synthetic electric trucks on long motorway origin–destination pairs; range binds through a reduced initial state of charge, since if agents never need to charge, moving a charger changes nothing. A synthetic two-corridor grid, built to vary geometry under control, carries 150 trucks and two chargers; there range binds through trip length. Unmet charging energy is priced into the score linearly, at a rate derived per scenario from the scoring function’s own time terms (0.20 score points per kWh in Brandenburg, 0.08 in the grid): without it, skipping a needed charge outscores taking it, and perturbed populations drift into degenerate non-charging equilibria. Every run is 300 iterations with innovation disabled at iteration 270 and the clock read to iteration 269; each treatment gets three cold and three warm seeds, with warm-started null controls as acceptance gates.

Perturbations. In Brandenburg, thirteen treatments: relocation of one charger at 5 to 80 km displacement, addition, removal, and multi-charger variants, together orphaning 27 to 109 of the converged plans’ charging stops. In the grid, a 4×4 factorial of 16 treatments, crossing the displacement of one charger (5, 15, 40, 80 km) with four positions of the second, so that the spacing between the two ranges from 10 to 160 km; displacement and spacing varied near-independently (correlation -0.283).

Measurement. At least four convergence criteria are in active use in MATSim work: the per-agent equilibrium gap of Flötteröd’s *greedo*, the executed-score criterion of Zhuge et al. (2021), a smoothed mode-share derivative (Hörl 2022), and KPSS with ADF stationarity tests (Koutounidis et al. 2025). A reconvergence time is criterion-dependent, so results can only be compared across studies when the same convergence criterion is used; ours is therefore stated in full. Reconvergence is read against the perturbed layout’s own equilibrium, estimated by the band its own cold runs settle into, because perturbed layouts are not equally good and several end above the baseline. The score clock is the first iteration from which the warm run’s best-plan score (*avg_best*) remain above its cold band’s floor (Figure 1); per-KPI clocks, on charging starts, queueing time, detour ratio, vehicle kilometres and energy delivered, are two-sided against their own cold bands, since required tolerance is KPI-dependent (Patil et al. 2021). An executed-score criterion such as Zhuge’s is not usable in this replanning regime: with innovation active, roughly a tenth of agents execute a fresh plan each iteration, holding the executed score about two points below the best, so its flattening detects the innovation shutoff rather than equilibrium. Following Boyce et al. (2004), error is measured from the difference between scenarios, which is all an outer loop consumes. A clock of zero is censoring, not measurement: the warm run remains within its reference band. Censored treatments are excluded from fits and their count reported.

What the measurement shows. Reconvergence cost is governed by the quality change of the perturbed configuration rather than by the magnitude of the perturbation, over the displacements tested and in the two scenarios examined. In Brandenburg, the seven treatments orphaning the same 27 stops share an identical iteration-0 deficit of 6.08 score points, the converged plans re-scored against the new layout before any replanning, and take 26 to 208 iterations: one value of the deficit, an eightfold spread in the response. In the grid factorial (Figure 2), the two designed axes together explain almost none of the clock (R^2 0.012 over the 13 of 16 treatments with a measured time, 3 left-censored), while the perturbed layout’s own equilibrium quality, its cold floor minus its baseline’s, orders it (adjusted R^2 0.252 alone, 0.363 with all three predictors). The coefficient is positive: layouts with higher equilibrium scores take longer reconvergence times. Absorbing a degradation asks nothing of the population; realising an improvement asks it to find the better-placed charger and re-route to it. The clock here measures the time to find a gain, not the time to repair damage. For the mechanisms above, this is a constraint: truncation bias cannot be scheduled on move size, and the one variable observed to govern the cost is an outcome of the very run being priced. For the outer loop, it is a warning: every charger proposed by a siting optimiser is intended as an improvement, so the most favored moves of the outer search are the ones that cost most to evaluate.

The censored treatments carry a second result. The three grid treatments whose score clock reads zero never fell below their floors, yet end outside their own cold bands on charging starts and unmet energy: the score declares immediate reconvergence for treatments that never reconverged physically. Before unmet energy entered the score, seven of the eleven

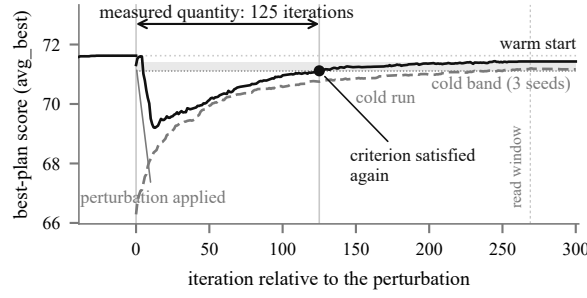


Figure 1: The measured quantity, on one run of the graded Brandenburg matrix (all three chargers relocated by 15 km, seed 4711): iterations from the applied perturbation until the warm-started run stays inside the perturbed layout’s own cold seed band, here 125, with a cold run on the same perturbed scenario for comparison.

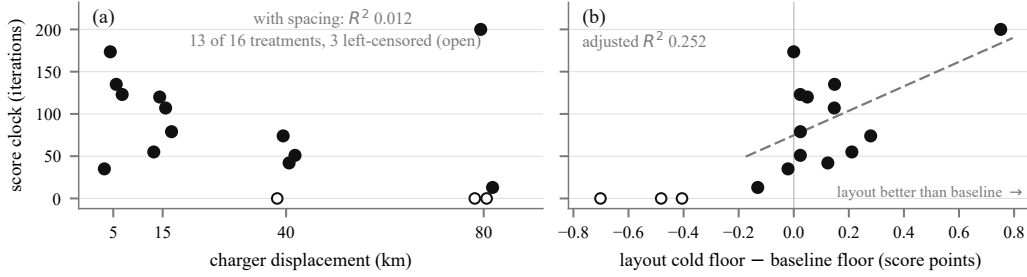


Figure 2: The grid factorial’s clocks (per-treatment medians over three seeds) against (a) the designed displacement axis and (b) the perturbed layout’s own quality change. Open markers are the three left-censored treatments, whose warm runs never left their reference bands; they are excluded from both fits. The dashed line is the fit over the 13 measurable treatments.

disrupted Brandenburg treatments were censored this way; after, one. Charging is the probe rather than the subject: it carries an independent physical observable, i.e., the consumed fleet energy, against which the score can be checked, and a scenario without such an observable would report those zeros as measurements.

5 Outlook

Reconvergence cost is governed by the quality change a move produces, and estimating a configuration’s quality without simulating it is what surrogate models in this lineage already do (Patwary et al. 2021; Flötteröd 2017). The open question this result poses is whether a surrogate’s *predicted* improvement can price an evaluation before it is paid for: an outer loop that knows a candidate is a large improvement can budget inner iterations for it, or defer it, rather than discovering the cost after spending it. The co-design corollary survives in that changed form: the quantity that prices an inner evaluation is one the outer level is already estimating, not the size of the move it contemplates. We will present the full run matrices and the per-KPI reconvergence curves at the meeting.

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